

UTS Cloud Computing Class - Lab Guide

Week 7 Lab: Resources and HA

This week continues configuring and setting up different resource and permission mechanisms in a vSphere environment. These features are important in order to create a viable multi-tenancy solution that may involve different tenants receiving different permissions, and resource allocations.

Task 1. Prepare for the lab.

1. Open VMware Workstation, and **power on** the following five/six VMs: 'ESXi01', 'ESXi02', 'VCSA' + 'OpenWRT' and 'iSCSI' ('DT' for F2F students).
2. Verify that all your machines are online and connected to the network.
3. Connect to your VCSA vm using a **web browser** (from 'DT') (<https://172.20.10.94>).
4. Power up the VMs that you have created using OVF on your ESXi hosts.

** **STOP** * Ping from the Terminal (or cmd) window to check if there is a connection to ESXi01 and ESXi02. If connectivity is not established follow the steps outlined from earlier weeks before continuing.*

Task 2. Manage a Cluster.

Last week we created resource pools on hosts. They are effective segmentations of hosts underlying hardware capacity (cpu/memory). When we created a Cluster object, this became an aggregation of resources from multiple hosts. This effectively abstracts the performance of the VM away from any one individual host, and places that responsibility with the Cluster.

NOTE The **Scheduling Affinity** setting for the virtual machines should be removed before moving the ESXi hosts to the cluster.

1. Select Your **Datacenter**.
2. Verify your new **Cluster** called *YourName-Cluster* has two hosts and several VMs.
3. Drag ESXi 2 out of the cluster.
4. What happened? _____
5. Notice that your VMs are no longer displayed under each **host** but are alongside the hosts in the **cluster**. This signifies that the VMs are being serviced by the cluster, and that the hosts service the cluster.
6. Create a **Resource Pool** in the cluster called "Pool1" and assign a cpu limit of 500Mhz. Leave the Share value as default.

***Hint:* If the 'New Resource Pool' option is grayed out on VCSA, enable vSphere DRS setting on the Cluster. Click on Cluster -> Configure tab -> Under Services, vSphere DRS -> Edit -> Turn ON vSphere DRS

7. Try creating the Resource Pool again and add any VM (OVF or clone) to the resource pool and run the **cpubusy.sh** script like last time.
8. **Migrate** your VM to the other host.
9. Notice that it will not matter what host the vm is running on, it will always be limited to 500Mhz cpu.

10. Your cluster should have 2 hosts, and at least 2 VMs.

11. Make sure the VMs are powered

Task 3. Enable HA 'High Availability' is the VMware product that allows for VMs to be restarted if the host that they reside on has failed. The system can automatically power-up VMs that were running onto another host in the cluster. This is a cluster feature. it cannot work with standalone hosts.

1. Go to the settings of your **Cluster** on VCSA.
2. Enable the setting for **HA** (below the vSphere DRS setting) and change **Heartbeat Datastores** to 'Automatically select datastore accessible from the host'.
3. Add configuration parameters and click **OK**.
4. Notice in your tasks that it is enabling HA for each **host** and performing an election.
5. Once complete, navigate to the **Monitor** tab of the cluster on VCSA.
6. Under the **Monitor tab -> vSphere HA -> Summary**, find out which host is the **Master** host.
7. Record which host took that role _____.
8. All other hosts are **slave** hosts, there is only one master in a cluster.
9. **Migrate** some running VMs to the Slave host, including your first VM with the running script.

Task 4. Create a host failure event.

We are going to simulate a host failure by rebooting a host that is currently running VMs. The VMs will lose power and be powered on again (on a different host) by HA automatically.

1. How many VMs are running on the **Master** Host? _____
2. How many VMs are running on the **Slave** Host? _____ 3. While there are VMs running on the **slave** host, select the host and shut it down.
4. **Observe** the events that occur in vCenter.
5. How many VMs are running on the **Master** Host now? _____
6. Did the VMs that were on the **slave** power up on the master host?
7. Verify that your host is now powered off in VMware Workstation, but the VMs are running on the other host.
8. Is your first VM still running the cpubusy.sh script?
9. Why / Why not? _____

NOTE These are warning messages and can be safely ignored. The warnings are because not all the VMware Tools are not installed for the virtual machines.

- ① The number of vSphere HA heartbeat datastores for this host is 1, which is less than required: 2
- ① Unable to apply DRS resource settings on host. . This can significantly reduce the effectiveness of DRS.

When a host is shutdown, the vCenter may report "the failover process failed". You may see the following warning:

"vSphere HA agent on this host could not reach isolation address '172.20.20.2'" reported on ESXi hosts.

The workaround is:

- a. Login to vCenter host (172.20.20.94) and set the **Default Gateway to 172.20.20.1**.
- b. Login to the hosts (172.20.20.51 and 172.20.20.52) and go to **Configure Management Network**, select **IPv4 Configuration**. Set the **Default Gateway to 172.20.20.1**.

c. Restart the ESXi hosts (if you don't restart the hosts, synchronization with vCenter may take a long time).

Task 5. Prepare for the next lab.

1. **Power off** your remaining VMs
2. **Shutdown** your other host.
3. **Shutdown** all other VMs in VMware Workstation.